

and tracheal intubation, during surgery, on emergence from anesthesia, and in the postoperative period, when in the physician’s judgment such specific intervention is considered indicated.
Use of ESOCARD INJECTION to prevent such events is not recommended.

Recommended Dose

Supraventricular Tachycardia

In the treatment of supraventricular tachycardia, responses to ESOCARD INJECTION usually (over 95%) occur within the range of 50 to 200 mcg/kg/min (0.05 to 0.2 mg/kg/mm). The average effective dosage is approximately 100 mcg/kg/min (0.1 mg/kg/mm) although dosages as low as 25 mcg/kg/min (0.025 mg/kg/mm) have been adequate in some patients. To initiate treatment of a patient with supraventricular tachycardia, administer a loading infusion of 500 mcg/kg/min (0.5 mg/kg/mm) over one minute followed by a four-minute maintenance infusion of 50 mcg/kg/min (0.05 mg/kg/mm). If an adequate therapeutic effect is observed over the five minutes of drug administration, maintain the maintenance infusion dosage with periodic adjustments up or down as needed. If an adequate therapeutic effect is not observed, the same loading dosage is repeated over one minute followed by an increased maintenance infusion rate of 100 mcg/kg/min (0.1 mg/kg/mm).

Continue titration procedure as above, repeating the original loading infusion of 500 mcg/kg/min (0.5 mg/kg/mm) over 1 minute, but increasing the maintenance infusion rate over the subsequent four minutes by 50 mcg/kg/min (0.05 mg/kg/mm) increments. As the desired heart rate or blood pressure is approached, omit subsequent loading doses and titrate the maintenance dosage up or down to endpoint. Also, if desired, increase the interval between steps from 5 to 10 minutes. The use of infusions of ESOCARD INJECTION up to 24 hours has been well documented; in addition, limited data from 24-48 hrs (N=48) indicate that ESOCARD INJECTION is well tolerated up to 48 hours.

Intraoperative and Postoperative Tachycardia and/or Hypertension

In the intraoperative and postoperative settings it is not always advisable to slowly titrate the dose of ESOCARD INJECTION to a therapeutic effect. Therefore, two dosing options are presented: immediate control dosing and a gradual control when the physician has time to titrate.

1. Immediate Control For intraoperative treatment of tachycardia and/or hypertension give an 80 mg (approximately 1 mg/kg) bolus dose over 30 seconds followed by a 150 mcg/kg/min infusion, if necessary. Adjust the infusion rate as required up to 300 mcg/kg/min to maintain desired heart rate and/or blood pressure.
2. Gradual Control For postoperative tachycardia and hypertension, the dosing schedule is the same as that used in supraventricular tachycardia. To initiate treatment, administer a loading dosage infusion of 500 mcg/kg/min of ESOCARD INJECTION for one minute followed by a fourminute maintenance infusion of 50 mcg/kg/min. If an adequate therapeutic effect is not observed within five minutes, repeat the same loading dosage and follow with a maintenance infusion increased to 100 mcg/kg/min .

Compatibility with Commonly Used Intravenous Fluids ESOCARD INJECTION was tested for compatibility with ten commonly used intravenous fluids at a final concentration of 10 mg esmolol HCl per mL. ESOCARD INJECTION was found to be compatible with the following solutions and was stable for at least 24 hours at controlled room temperature or under refrigeration:

- Dextrose (5%) Injection, USP
- Dextrose (5%) in Lactated Ringer’s Injection
- Dextrose (5%) in Ringer’s Injection
- Dextrose (5%) and Sodium Chloride (0.45%) Injection, USP
- Dextrose (5%) and Sodium Chloride (0.9%) Injection, USP
- Lactated Ringer’s Injection, USP
- Potassium Chloride (40 mEq/liter) in Dextrose (5%) Injection, USP
- Sodium Chloride (0.45%) Injection, USP
- Sodium Chloride (0.9%) Injection, USP
- ESOCARD INJECTION was NOT compatible with Sodium Bicarbonate (5%) Injection, USP.

Mode Of Administration

INTRAVENOUS INJECTION

Contraindication

ESOCARD INJECTION is contraindicated in patients with sinus bradycardia, heart block greater than first degree, cardiogenic shock or overt heart failure .

A10. Warning and Precautions

Hypotension

Hypotension can occur at any dose but is dose-related so that doses beyond 200 mcg/kg/min (0.2 mg/kg/mm)

are not recommended. Patients should be closely monitored, especially if pretreatment blood pressure is low. Decrease of dose or termination of infusion reverses hypotension, usually within 30 minutes.

Cardiac Failure

Sympathetic stimulation is necessary in supporting circulatory function in congestive heart failure, and beta blockade carries the potential hazard of further depressing myocardial contractility and precipitating more severe failure. Continued depression of the myocardium with beta blocking agents over a period of time can, in some cases, lead to cardiac failure. At the first sign or symptom of impending cardiac failure, ESOCARD (esmolol HCl) should be withdrawn. The use of ESOCARD for control of ventricular response in patients with supraventricular arrhythmias should be undertaken with caution when the patient is compromised hemodynamically or is taking other drugs that decrease any or all of the following: peripheral resistance, myocardial filling, myocardial contractility, or electrical impulse propagation in the myocardium. Despite the rapid onset and offset of the effects of ESOCARD, several cases of death have been reported in complex clinical states where ESOCARD was presumably being used to control ventricular rate.

Intraoperative and Postoperative Tachycardia and/or Hypertension

ESOCARD (esmolol HCl) should not be used as the treatment for hypertension in patients in whom the increased blood pressure is primarily due to the vasoconstriction associated with hypothermia.

Bronchospastic Diseases

PATIENTS WITH BRONCHOSPASTIC DISEASES SHOULD, IN GENERAL, NOT RECEIVE BETA BLOCKERS. Because of its relative beta 1 selectivity and titratability, ESOCARD (esmolol HCl) may be used with caution in patients with bronchospastic diseases.

Diabetes Mellitus and Hypoglycemia

ESOCARD (esmolol HCl) should be used with caution in diabetic patients requiring a beta blocking agent. Beta blockers may mask tachycardia occurring with hypoglycemia, but other manifestations such as dizziness and sweating may not be significantly affected.

General

Because the acid metabolite of ESOCARD is primarily excreted unchanged by the kidney, ESOCARD (esmolol HCl) should be administered with caution to patients with impaired renal function. The elimination half-life of the acid metabolite was prolonged ten-fold and the plasma level was considerably elevated in patients with end stage renal disease. Care should be taken in the intravenous administration of ESOCARD as sloughing of the skin and necrosis have been reported in association with infiltration and extravasation of intravenous infusions.

Interactions With Other Medicaments

Catecholamine-depleting drugs, e.g., reserpine, may have an additive effect when given with beta blocking agents. Patients treated concurrently with ESOCARD INJECTION (esmolol HCl) and a catecholamine depletor should therefore be closely observed for evidence of hypotension or marked bradycardia, which may result in vertigo, syncope, or postural hypotension. A study of interaction between ESOCARD INJECTION and warfarin showed that concomitant administration of ESOCARD INJECTION and warfarin does not alter warfarin plasma levels. When digoxin and ESOCARD INJECTION were concomitantly administered intravenously to normal volunteers, there was a 10-20% increase in digoxin blood levels at some time points. Digoxin did not affect ESOCARD INJECTION pharmacokinetics. When intravenous morphine and ESOCARD INJECTION were concomitantly administered in normal subjects, no effect on morphine blood levels was seen, but ESOCARD INJECTION steady-state blood levels were increased by 46% in the presence of morphine. No other pharmacokinetic parameters were changed. The effect of ESOCARD INJECTION on the duration of succinylcholine-induced neuromuscular blockade was studied in patients undergoing surgery. The onset of neuromuscular blockade by succinylcholine was unaffected by ESOCARD INJECTION, but the duration of neuromuscular blockade was prolonged from 5 minutes to 8 minutes. Caution should be exercised when considering the use of ESOCARD INJECTION and verapamil in patients with depressed myocardial function. Fatal cardiac arrests have occurred in patients receiving both drugs. Additionally, ESOCARD INJECTION should not be used to control supraventricular tachycardia in the presence of agents which are vasoconstrictive and inotropic such as dopamine, epinephrine, and norepinephrine because of the danger of blocking cardiac contractility when systemic vascular resistance is high. **Pregnancy and Lactation**

Pregnancy Category

Although there are no adequate and well-controlled studies in pregnant women, use of esmolol in the last trimester of pregnancy or during labor or delivery has been reported to cause fetal bradycardia, which continued after termination of drug infusion. ESOCARD INJECTION should be used during pregnancy only if the potential

ESOCARD INJECTION 10 MG / ML

Name and Strength of Active Ingredient (s)

No.	Substance	Quantity Unit Per Dose
1.	ESMOLOL HYDROCHLORIDE	10.0 mg; 1.0 ml

Dosage Form : Parenteral Injection, liquid

Product Description

Clear colorless solution filled in 10 ml amber vial plugged with 13 mm grey butyl rubber plug & sealed with 13 mm green colored flip off Aluminium seals. 1 vial in a carton with leaflet.

Pharmacodynamic

ESOCARD (esmolol HCl) is a beta1-selective (cardioselective) adrenergic receptor blocking agent with rapid onset, a very short duration of action, and no significant intrinsic sympathomimetic or membrane stabilizing activity at therapeutic dosages. Its elimination half life after intravenous infusion is approximately 9 minutes. ESOCARD inhibits the beta1 receptors located chiefly in cardiac muscle, but this preferential effect is not absolute and at higher doses it begins to inhibit beta2 receptors located chiefly in the bronchial and vascular musculature.

Pharmacokinetics

Pharmacokinetics and Metabolism

Esocard (Esmolol Hydrochloride) is rapidly metabolized by hydrolysis of the ester linkage, chiefly by the esterases in the cytosol of red blood cells and not by plasma cholinesterases or red cell membrane acetylcholinesterase. Total body clearance in man was found to be about 20 L/kg/hr, which is greater than cardiac output; thus the metabolism of Esocard is not limited by the rate of blood flow to metabolizing tissues such as the liver or affected by hepatic or renal blood flow. Esocard has a rapid distribution half life of about 2 minutes and an elimination half-life of about 9 minutes.

Using an appropriate loading dose, steady state blood levels of Esocard for dosages from 50-300 mcg/kg/min (0.05-0.3 mg/kg/min) are obtained within five minutes. (Steady-state is reached in about 30 minutes without the loading dose.) Steady-state blood levels of Esocard increase linearly over this dosage range and elimination kinetics are dose independent over this range. Steady-state blood levels are maintained during infusion but decrease rapidly after termination of the infusion. Because of its short half-life, blood levels of Esocard can be rapidly altered by increasing or decreasing the infusion rate and rapidly eliminated by discontinuing the infusion. Consistent with the high rate of blood-based metabolism of Esocard, less than 2% of the drug is excreted unchanged in the urine. Within 24 hours of the end of infusion, approximately 73-88% of the dosage has been accounted for in the urine as the acid metabolite of Esocard.

Metabolism of Esocard results in the formation of the corresponding free acid and methanol. The acid metabolite has been shown in animals to have about 1/1500th the activity of esmolol and in normal volunteers its blood levels do not correspond to the level of beta blockade. The acid metabolite has an elimination half-life of about 3.7 hours and is excreted in the urine with a clearance approximately equivalent to the glomerular filtration rate. Excretion of the acid metabolite is significantly decreased in patients with renal disease, with the elimination half-life increased to about ten-fold that of normals, and plasma levels considerably elevated. Methanol blood levels, monitored in subjects receiving Esocard for up to 6 hours at 300 mcg/kg/min (0.3 mg/kg/min) and 24 hours at 150 mcg/kg/min (0.15 mg/kg/min), approximated endogenous levels and were less than 2% of levels usually associated with methanol toxicity. Esocard has been shown to be 55% bound to human plasma protein, while the acid metabolite is only 10% bound.

Indication

Supraventricular Tachycardia

ESOCARD INJECTION is indicated for the rapid control of ventricular rate in patients with atrial fibrillation or atrial flutter in perioperative, postoperative, or other emergent circumstances where short term control of ventricular rate with a short-acting agent is desirable. ESOCARD INJECTION is also indicated in noncompensatory sinus tachycardia where, in the physician’s judgment, the rapid heart rate requires specific intervention.

Intraoperative and Postoperative Tachycardia and/or Hypertension

ESOCARD INJECTION is indicated for the treatment of tachycardia and hypertension that occur during induction

benefit justifies the potential risk to the fetus.

Nursing Mothers

It is not known whether ESOCARD INJECTION is excreted in human milk; however, caution should be exercised when ESOCARD INJECTION is administered to a nursing woman.

Pediatric Use

The safety and effectiveness of ESOCARD INJECTION in children have not been established.

Side Effects

Most adverse effects observed in controlled clinical trial settings have been mild and transient. The most important adverse effect has been hypotension. Deaths have been reported in postmarketing experience occurring during complex clinical states where ESOCARD INJECTION was presumably being used simply to control ventricular rate . **Cardiovascular** : Symptomatic hypotension (diaphoresis, dizziness), Asymptomatic hypotension. Diaphoresis accompanied hypotension .Peripheral ischemia ,Pallor, flushing, bradycardia (heart rate less than 50 beats per minute), chest pain, syncope, pulmonary edema and heart block have each been reported. In two patients without supraventricular tachycardia but with serious coronary artery disease (post inferior myocardial infarction or unstable angina), severe bradycardia/sinus pause/asystole has developed, reversible in both cases with discontinuation of treatment.

Central Nervous System : Dizziness ,somnolence, headache, and agitation and fatigue . Paresthesia, asthenia, depression, abnormal thinking, anxiety, anorexia, and lightheadedness were also reported .Seizures were also reported. **Respiratory** : Bronchospasm, wheezing, dyspnea, nasal congestion, rhonchi, and rales have each been reported . **Gastrointestinal** – Nausea,Vomiting, Dyspepsia, constipation, dry mouth, and abdominal discomfort have occurred. Taste perversion has also been reported. Skin (Infusion Site) - Infusion site reactions including inflammation and induration were reported .Edema, erythema, skin discoloration, burning at the infusion site, thrombophlebitis, and local skin necrosis from extravasation have occurred .

Miscellaneous : Urinary retention, speech disorder, abnormal vision, midscapular pain, rigors, and fever.

Symptoms And Treatment of Overdose Acute Toxicity

Overdoses of ESOCARD INJECTION can cause cardiac arrest. In addition, overdoses can produce bradycardia, hypotension, electromechanical dissociation and loss of consciousness. Some of these overdoses have been fatal while others resulted in permanent disability. Based on the observed clinical effects, the following general measures should also be considered.

Bradycardia: Intravenous administration of atropine or another anticholinergic drug.

Bronchospasm: Intravenous administration of a beta2 stimulating agent and/or a theophylline derivative.

Cardiac Failure: Intravenous administration of a diuretic and/or digitalis glycoside. In shock resulting from inadequate cardiac contractility, intravenous administration of dopamine, dobutamine, isoproterenol, or amrinone may be considered.

Symptomatic Hypotension: Intravenous administration of fluids and/or pressor agents.

Storage Condition

Store at a temperature below 30°C. Protect from light.Do not freeze.

Shelf Life

24 Months.

Presentation: 1vial of 10 ml.

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Manufactured by :



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Product Registration Holder

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